

Assignment 3.1 Exercise

AAI-531 | Module 3

In this assignment, you will explore the impact of fairness metrics after applying the Disparate Impact Remover (DIR), retrain a Light GBM model with the adjusted data, and compare its metrics (listed below) against the original and reweighted models from the lab tutorial. Afterward, respond to the reflection questions.

Part 1: Experimentation with the DIR

- Modify the levels array for the Disparate Impact Remover to include finer and coarser granularity. (See example in lab/assignment notebook.)
- Identify the best repair level and the Disparate Impact at the best repair level.
- Retrain a Light GBM model on the modified data from the DIR and compute both the performance and fairness metrics.

1.1a Code Output for Finer Granularity

Paste your screenshot here. *(This can be a very lengthy output)*

Attach a screenshot or output of your code here:

1%| | 1/101 [00:03<05:55, 3.55s/it]Repair Level: 0.00, Disparate Impact: 0.8350

2%|▌ | 2/101 [00:07<05:45, 3.49s/it]Repair Level: 0.00, Disparate Impact: 0.8434

3%|▊ | 3/101 [00:10<05:54, 3.62s/it]Repair Level: 0.00, Disparate Impact: 0.8464

4%|▉ | 4/101 [00:14<05:49, 3.60s/it]Repair Level: 0.00, Disparate Impact: 0.8467

5%|▉ | 5/101 [00:17<05:42, 3.57s/it]Repair Level: 0.01, Disparate Impact: 0.8401

6%|▏ | 6/101 [00:21<05:42, 3.61s/it]Repair Level: 0.01, Disparate Impact: 0.8448

7%|▍ | 7/101 [00:25<05:37, 3.59s/it]Repair Level: 0.01, Disparate Impact: 0.8405

8%|▋ | 8/101 [00:28<05:34, 3.59s/it]Repair Level: 0.01, Disparate Impact: 0.8367

9%|▊ | 9/101 [00:32<05:29, 3.58s/it]Repair Level: 0.01, Disparate Impact: 0.8419

10%|▉ | 10/101 [00:35<05:24, 3.56s/it]Repair Level: 0.01, Disparate Impact: 0.8456

11%|▉ | 11/101 [00:39<05:21, 3.57s/it]Repair Level: 0.01, Disparate Impact: 0.8330

12%|▉ | 12/101 [00:42<05:17, 3.57s/it]Repair Level: 0.01, Disparate Impact: 0.8420

13%|▉ | 13/101 [00:46<05:10, 3.53s/it]Repair Level: 0.01, Disparate Impact: 0.8332

14%|▉ | 14/101 [00:49<05:06, 3.52s/it]Repair Level: 0.02, Disparate Impact: 0.8411

15%|▉ | 15/101 [00:53<05:02, 3.51s/it]Repair Level: 0.02, Disparate Impact: 0.8338

16%|▉ | 16/101 [00:57<05:06, 3.61s/it]Repair Level: 0.02, Disparate Impact: 0.8443

17%|▉ | 17/101 [01:00<05:01, 3.59s/it]Repair Level: 0.02, Disparate Impact: 0.8365

18%|▉ | 18/101 [01:04<04:53, 3.54s/it]Repair Level: 0.02, Disparate Impact: 0.8367

19%|▉ | 19/101 [01:07<04:47, 3.51s/it]Repair Level: 0.02, Disparate Impact: 0.8393

20%|▉ | 20/101 [01:11<04:42, 3.49s/it]Repair Level: 0.02, Disparate Impact: 0.8381

21%|▉ | 21/101 [01:14<04:40, 3.51s/it]Repair Level: 0.03, Disparate Impact: 0.8398

22%|▉ | 22/101 [01:18<04:38, 3.52s/it]Repair Level: 0.03, Disparate Impact: 0.8440

23%|▉ | 23/101 [01:21<04:32, 3.49s/it]Repair Level: 0.03, Disparate Impact: 0.8317

24%|▉ | 24/101 [01:25<04:34, 3.57s/it]Repair Level: 0.03, Disparate Impact: 0.8336

25%|▉ | 25/101 [01:28<04:29, 3.54s/it]Repair Level: 0.03, Disparate Impact: 0.8422

26%|▉ | 26/101 [01:32<04:26, 3.55s/it]Repair Level: 0.03, Disparate Impact: 0.8390

27%|▉ | 27/101 [01:35<04:22, 3.55s/it]Repair Level: 0.03, Disparate Impact: 0.8403

28%|▉ | 28/101 [01:39<04:17, 3.52s/it]Repair Level: 0.03, Disparate Impact: 0.8367

29%|▉ | 29/101 [01:42<04:14, 3.53s/it]Repair Level: 0.04, Disparate Impact: 0.8363

30%|▉ | 30/101 [01:46<04:09, 3.51s/it]Repair Level: 0.04, Disparate Impact: 0.8459

31%|▉ | 31/101 [01:49<04:04, 3.50s/it]Repair Level: 0.04, Disparate Impact: 0.8439

32%|▉ | 32/101 [01:53<04:04, 3.55s/it]Repair Level: 0.04, Disparate Impact: 0.8439

33%|▉ | 33/101 [01:57<04:05, 3.61s/it]Repair Level: 0.04, Disparate Impact: 0.8417

34%|▉ | 34/101 [02:00<03:59, 3.57s/it]Repair Level: 0.04, Disparate Impact: 0.8429

35%|▉ | 35/101 [02:04<03:54, 3.55s/it]Repair Level: 0.04, Disparate Impact: 0.8447

36%|████ | 36/101 [02:07<03:48, 3.52s/it]Repair Level: 0.04, Disparate Impact: 0.8400

37%|████▏ | 37/101 [02:11<03:48, 3.57s/it]Repair Level: 0.04, Disparate Impact: 0.8406

38%|████▏ | 38/101 [02:14<03:41, 3.52s/it]Repair Level: 0.05, Disparate Impact: 0.8307

39%|████▏ | 39/101 [02:18<03:36, 3.49s/it]Repair Level: 0.05, Disparate Impact: 0.8439

40%|████▏ | 40/101 [02:21<03:33, 3.50s/it]Repair Level: 0.05, Disparate Impact: 0.8368

41%|████▏ | 41/101 [02:25<03:30, 3.51s/it]Repair Level: 0.05, Disparate Impact: 0.8377

42%|████▎ | 42/101 [02:28<03:26, 3.50s/it]Repair Level: 0.05, Disparate Impact: 0.8338

43%|████▎ | 43/101 [02:32<03:24, 3.52s/it]Repair Level: 0.05, Disparate Impact: 0.8327

44%|████▎ | 44/101 [02:35<03:20, 3.52s/it]Repair Level: 0.05, Disparate Impact: 0.8366

45%|████▎ | 45/101 [02:39<03:16, 3.51s/it]Repair Level: 0.06, Disparate Impact: 0.8407

46%|████▍ | 46/101 [02:42<03:11, 3.49s/it]Repair Level: 0.06, Disparate Impact: 0.8358

47%|████▍ | 47/101 [02:46<03:07, 3.48s/it]Repair Level: 0.06, Disparate Impact: 0.8362

48%|████▍ | 48/101 [02:49<03:05, 3.50s/it]Repair Level: 0.06, Disparate Impact: 0.8370

49%|████▍ | 49/101 [02:53<03:02, 3.50s/it]Repair Level: 0.06, Disparate Impact: 0.8369

50%|████▌ | 50/101 [02:56<03:02, 3.57s/it]Repair Level: 0.06, Disparate Impact: 0.8404

50%|████▌ | 51/101 [03:00<02:57, 3.55s/it]Repair Level: 0.06, Disparate Impact: 0.8259

51%|████▌ | 52/101 [03:03<02:53, 3.54s/it]Repair Level: 0.06, Disparate Impact: 0.8374

52%|████▋ | 53/101 [03:07<02:50, 3.55s/it]Repair Level: 0.07, Disparate Impact: 0.8339

53%|████▋ | 54/101 [03:11<02:46, 3.53s/it]Repair Level: 0.07, Disparate Impact: 0.8376

54%|████▋ | 55/101 [03:14<02:40, 3.49s/it]Repair Level: 0.07, Disparate Impact: 0.8403

55%|████▊ | 56/101 [03:17<02:37, 3.51s/it]Repair Level: 0.07, Disparate Impact: 0.8370

56%|████▊ | 57/101 [03:21<02:34, 3.52s/it]Repair Level: 0.07, Disparate Impact: 0.8337

57%|████▊ | 58/101 [03:25<02:31, 3.53s/it]Repair Level: 0.07, Disparate Impact: 0.8298

58%|████▊ | 59/101 [03:28<02:27, 3.51s/it]Repair Level: 0.07, Disparate Impact: 0.8325

59%|████▉ | 60/101 [03:32<02:23, 3.50s/it]Repair Level: 0.07, Disparate Impact: 0.8372

60%|████▉ | 61/101 [03:35<02:19, 3.49s/it]Repair Level: 0.07, Disparate Impact: 0.8327

61%|██████ | 62/101 [03:38<02:15, 3.49s/it]Repair Level: 0.08, Disparate Impact: 0.8372

62%|██████ | 63/101 [03:42<02:11, 3.46s/it]Repair Level: 0.08, Disparate Impact: 0.8477

63%|██████ | 64/101 [03:45<02:09, 3.49s/it]Repair Level: 0.08, Disparate Impact: 0.8378

64%|██████ | 65/101 [03:49<02:04, 3.47s/it]Repair Level: 0.08, Disparate Impact: 0.8339

65%|██████ | 66/101 [03:52<02:02, 3.50s/it]Repair Level: 0.08, Disparate Impact: 0.8383

66%|██████ | 67/101 [03:56<02:01, 3.57s/it]Repair Level: 0.08, Disparate Impact: 0.8285

67%|██████ | 68/101 [04:00<01:56, 3.53s/it]Repair Level: 0.08, Disparate Impact: 0.8376

68%|██████ | 69/101 [04:03<01:54, 3.57s/it]Repair Level: 0.09, Disparate Impact: 0.8313

69%|██████ | 70/101 [04:07<01:49, 3.54s/it]Repair Level: 0.09, Disparate Impact: 0.8313

70%|██████ | 71/101 [04:10<01:45, 3.52s/it]Repair Level: 0.09, Disparate Impact: 0.8284

71%|██████ | 72/101 [04:14<01:42, 3.52s/it]Repair Level: 0.09, Disparate Impact: 0.8357

72%|██████ | 73/101 [04:17<01:37, 3.50s/it]Repair Level: 0.09, Disparate Impact: 0.8402

73%|██████ | 74/101 [04:21<01:33, 3.47s/it]Repair Level: 0.09, Disparate Impact: 0.8389

74%|██████ | 75/101 [04:24<01:31, 3.51s/it]Repair Level: 0.09, Disparate Impact: 0.8428

75%|██████ | 76/101 [04:28<01:27, 3.50s/it]Repair Level: 0.09, Disparate Impact: 0.8341

76%|██████ | 77/101 [04:31<01:24, 3.50s/it]Repair Level: 0.10, Disparate Impact: 0.8385

77%|██████ | 78/101 [04:35<01:20, 3.49s/it]Repair Level: 0.10, Disparate Impact: 0.8407

78%|██████ | 79/101 [04:38<01:17, 3.51s/it]Repair Level: 0.10, Disparate Impact: 0.8421

79%|██████ | 80/101 [04:42<01:13, 3.51s/it]Repair Level: 0.10, Disparate Impact: 0.8347

80%|██████ | 81/101 [04:45<01:10, 3.52s/it]Repair Level: 0.10, Disparate Impact: 0.8421

81%|██████ | 82/101 [04:49<01:07, 3.54s/it]Repair Level: 0.20, Disparate Impact: 0.8495

82%|██████ | 83/101 [04:52<01:03, 3.53s/it]Repair Level: 0.24, Disparate Impact: 0.8494

83%|██████ | 84/101 [04:56<01:00, 3.56s/it]Repair Level: 0.28, Disparate Impact: 0.8433

84%|██████ | 85/101 [05:00<00:56, 3.56s/it]Repair Level: 0.33, Disparate Impact: 0.8539

85%|██████ | 86/101 [05:03<00:52, 3.53s/it]Repair Level: 0.37, Disparate Impact: 0.8566

86%|████████▋ | 87/101 [05:06<00:48, 3.50s/it]Repair Level: 0.41, Disparate Impact: 0.8597

87%|████████▊ | 88/101 [05:10<00:45, 3.50s/it]Repair Level: 0.45, Disparate Impact: 0.8605

88%|████████▉ | 89/101 [05:13<00:41, 3.49s/it]Repair Level: 0.49, Disparate Impact: 0.8659

89%|█████████ | 90/101 [05:17<00:38, 3.48s/it]Repair Level: 0.54, Disparate Impact: 0.7930

90%|█████████ | 91/101 [05:20<00:34, 3.46s/it]Repair Level: 0.58, Disparate Impact: 0.7894

91%|█████████▏ | 92/101 [05:24<00:30, 3.43s/it]Repair Level: 0.62, Disparate Impact: 0.7959

92%|█████████▎| 93/101 [05:27<00:26, 3.37s/it]Repair Level: 0.66, Disparate Impact: 0.8057

93%|█████████▍| 94/101 [05:30<00:23, 3.34s/it]Repair Level: 0.71, Disparate Impact: 0.8161

94%|█████████▍| 95/101 [05:33<00:19, 3.30s/it]Repair Level: 0.75, Disparate Impact: 0.8368

95%|█████████▌ | 96/101 [05:37<00:16, 3.30s/it]Repair Level: 0.79, Disparate Impact: 0.8335

96%|█████████▌ | 97/101 [05:40<00:12, 3.23s/it]Repair Level: 0.83, Disparate Impact: 0.8341

97%|█████████▋| 98/101 [05:43<00:09, 3.23s/it]Repair Level: 0.87, Disparate Impact: 0.8257

98%|█████████▊| 99/101 [05:46<00:06, 3.23s/it]Repair Level: 0.92, Disparate Impact: 0.8142

99%|█████████▊| 100/101 [05:49<00:03, 3.19s/it]Repair Level: 0.96, Disparate Impact: 0.8074

100%|██████████| 101/101 [05:52<00:00, 3.49s/it]Repair Level: 1.00, Disparate Impact: 0.8123

1.1b Code Output for Coarser Granularity

Paste your screenshot here.

7%|▉ | 1/15 [00:03<00:51, 3.67s/it]Repair Level: 0.00, Disparate Impact: 0.8350

13%|▉▉ | 2/15 [00:07<00:46, 3.56s/it]Repair Level: 0.01, Disparate Impact: 0.8333

20%|▉▉▉ | 3/15 [00:10<00:42, 3.51s/it]Repair Level: 0.02, Disparate Impact: 0.8353

27%|▉▉▉▉ | 4/15 [00:14<00:38, 3.53s/it]Repair Level: 0.03, Disparate Impact: 0.8438

33%|▉▉▉▉▉ | 5/15 [00:17<00:35, 3.54s/it]Repair Level: 0.04, Disparate Impact: 0.8379

40%|▉▉▉▉▉▉ | 6/15 [00:21<00:32, 3.63s/it]Repair Level: 0.06, Disparate Impact: 0.8334

47%|▉▉▉▉▉▉▉ | 7/15 [00:25<00:28, 3.60s/it]Repair Level: 0.07, Disparate Impact: 0.8262

53%|▉▉▉▉▉▉▉▉ | 8/15 [00:28<00:25, 3.60s/it]Repair Level: 0.08, Disparate Impact: 0.8464

60%|▉▉▉▉▉▉▉▉▉ | 9/15 [00:32<00:21, 3.58s/it]Repair Level: 0.09, Disparate Impact: 0.8356

67%|▉▉▉▉▉▉▉▉▉▉ | 10/15 [00:35<00:17, 3.53s/it]Repair Level: 0.10, Disparate Impact: 0.8421

73%|▉▉▉▉▉▉▉▉▉▉▉ | 11/15 [00:39<00:14, 3.55s/it]Repair Level: 0.20, Disparate Impact: 0.8495

80%|▉▉▉▉▉▉▉▉▉▉▉▉ | 12/15 [00:42<00:10, 3.52s/it]Repair Level: 0.40, Disparate Impact: 0.8683

87%|▉▉▉▉▉▉▉▉▉▉▉▉▉ | 13/15 [00:45<00:06, 3.45s/it]Repair Level: 0.60, Disparate Impact: 0.7952

93%|▉▉▉▉▉▉▉▉▉▉▉▉▉▉| 14/15 [00:49<00:03, 3.37s/it]Repair Level: 0.80, Disparate Impact: 0.8349

100%|▉▉▉▉▉▉▉▉▉▉▉▉▉▉▉| 15/15 [00:52<00:00, 3.48s/it]Repair Level: 1.00, Disparate Impact: 0.8123

1.2 Code Output for Best Repair Level and Disparate Impact at Best Repair Level

Paste your screenshot here. *(Choose the single best out of all repair levels for finer/coarser. This will be the DI closest to 1.0.)*

Coarser granularity

Best Repair Level: 0.40 Disparate Impact at Best Level: 0.8683

1.3 Code Output for Light GBM Performance Metrics and Fairness Metrics

Comparison of Performance Metrics Across Models

Metric	Original	Reweighted	DIR Applied
Accuracy	0.8176	0.8057	0.8109
Precision	0.6488	0.6140	0.6281
Recall	0.5404	0.5360	0.5404
F1-Score	0.5897	0.5724	0.5810
ROC-AUC	0.7991	0.7908	0.7949

Comparison of Fairness Metrics Across Models

Metric	Original	Reweighted	DIR Applied
SPD	-0.1385	-0.0541	-0.1078
DI	0.8339	0.9326	0.8683
SEDF	0.9115	0.7310	0.9115

***Comparison of Performance Metrics from the Original and Reweighted Models** *(for your convenience from the lab)*

Metric	Original	Reweighted
Accuracy	0.8176	0.8057
Precision	0.6488	0.6140
Recall	0.5404	0.5360
F1-Score	0.5897	0.5724
ROC-AUC	0.7991	0.7908

***Comparison of Fairness Metrics from the Original and Reweighted Models** *(for your convenience from the lab)*

Metric	Original	Reweighted
SPD	-0.1385	-0.0541
DI	0.8339	0.9326
SEDF	0.9115	0.7310

Part 2: Short-Response Questions

Question 2.1

How does the Disparate Impact (DI) change with finer granularity? Coarser granularity?

Your response: The DI for both finer and coarser granularity is higher than the baseline while the DI for both finer and coarser granularity are close to each other.

Question 2.2

Based on your findings, which model would you recommend (Original, Reweighted, or DIR)? Justify your choice by balancing performance and fairness.

Your response: I would recommend finer granularity since the best level was .50 at a similar DI to the actual best DI at the coarser granularity.

Question 2.3

Consider a scenario where you are responsible for deploying a machine learning model that shows a fairness-accuracy trade-off.

How would you approach deciding whether to prioritize fairness or accuracy? Which factors (organizational values, societal impact, or application context) would guide your decision? Reflect on how ethical dilemmas in machine learning require balancing competing values and their implications.

Your response: It depends on how badly a biased or inaccurate decision would affect the entity. If it could change someone's life and the harm is hard to undo then fairness should be prioritized and when there is little harm then accuracy would be prioritized.